average numbers over many repetitions: we want graphs to be drawn nicely, and there is no magic formula to measure “niceness”. The only way to judge the results is presenting them to the human eye.

Figure 17.17 is, I think, the perfect summary to explain what we have been building in these last couple of sections: we are trying to come up with a nice embedding for the square grid graph with side 4, a graph with 16 vertices arranged like a square mesh.

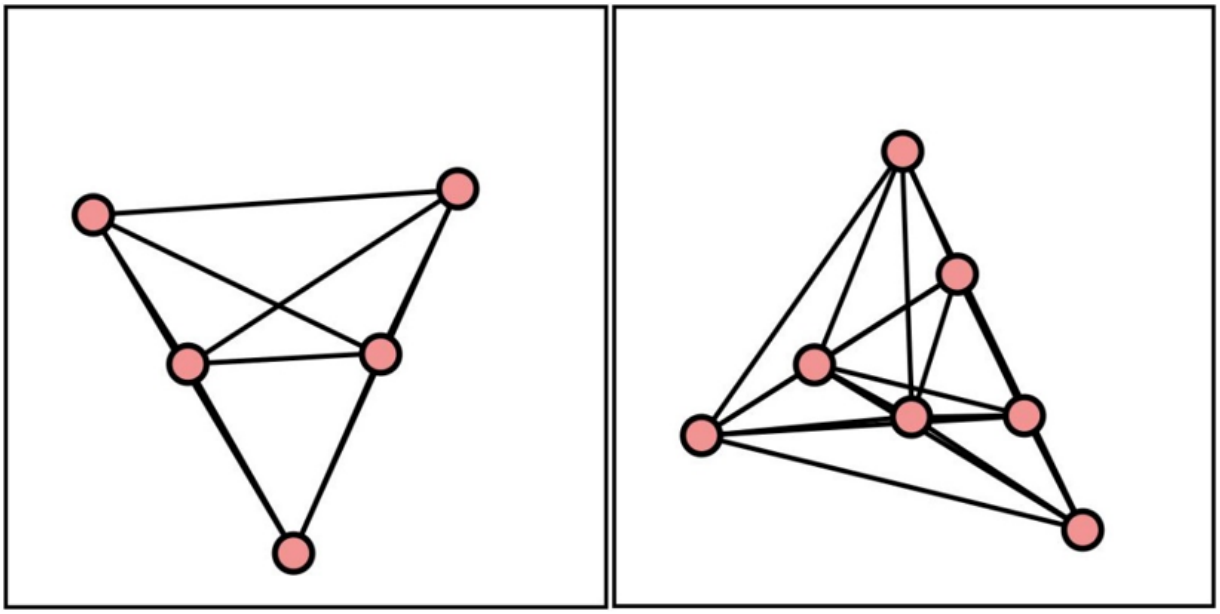


Figure 17.19 Complete graphs K_5 and K_7 embedded by Davidson and Harel’s algorithm. Here we could use that fifth component of the cost function keeping vertices away from edges…

Random sampling struggles even to find an embedding without intersections, goal that is reached by the algorithm presented in section 17.3.1, which, however, doesn’t do a